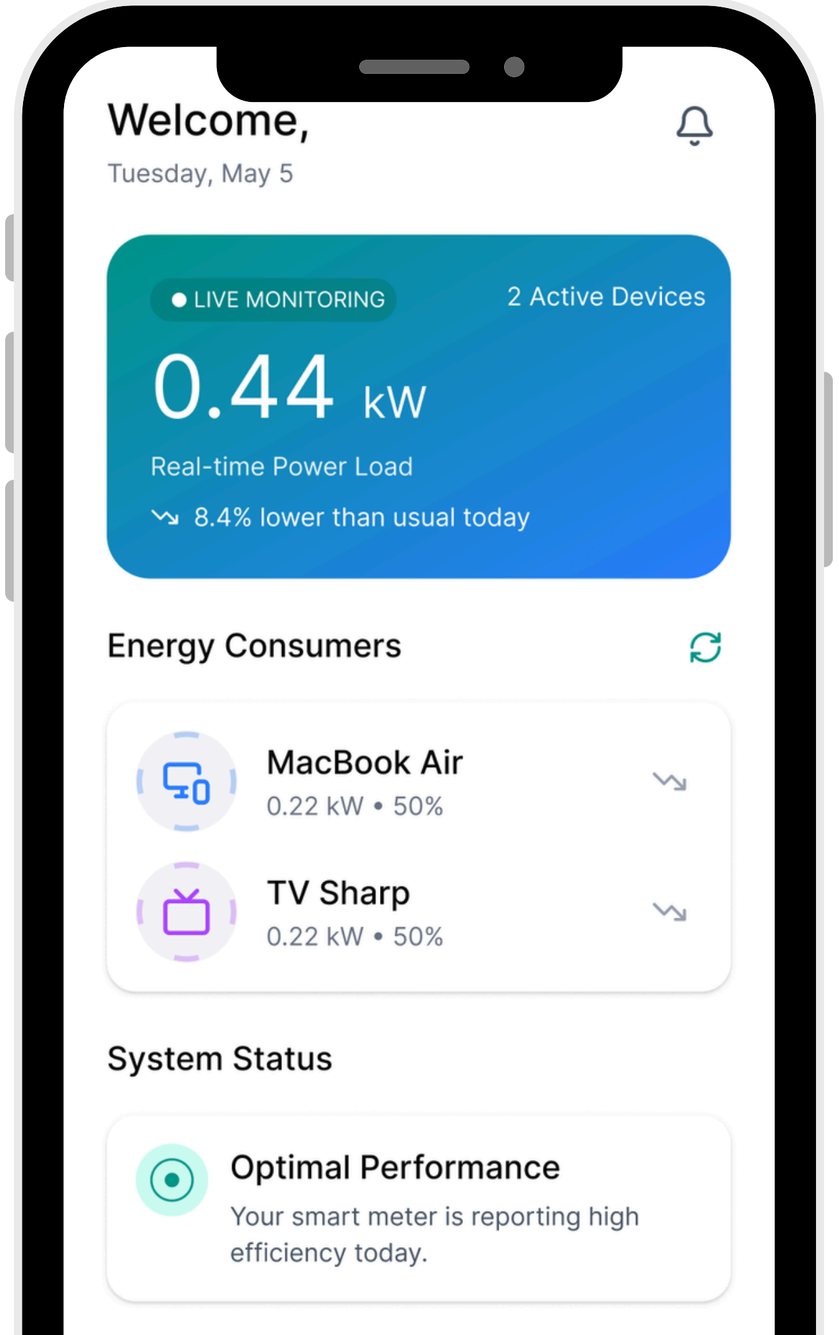
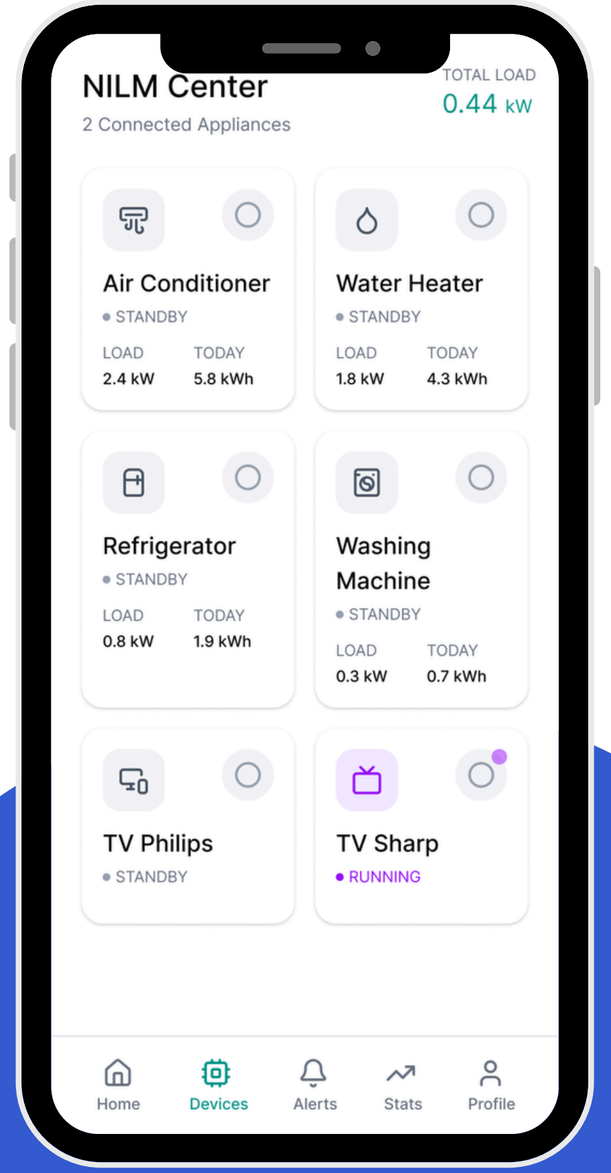
Smart Meter Monitoring System

Mobile APP



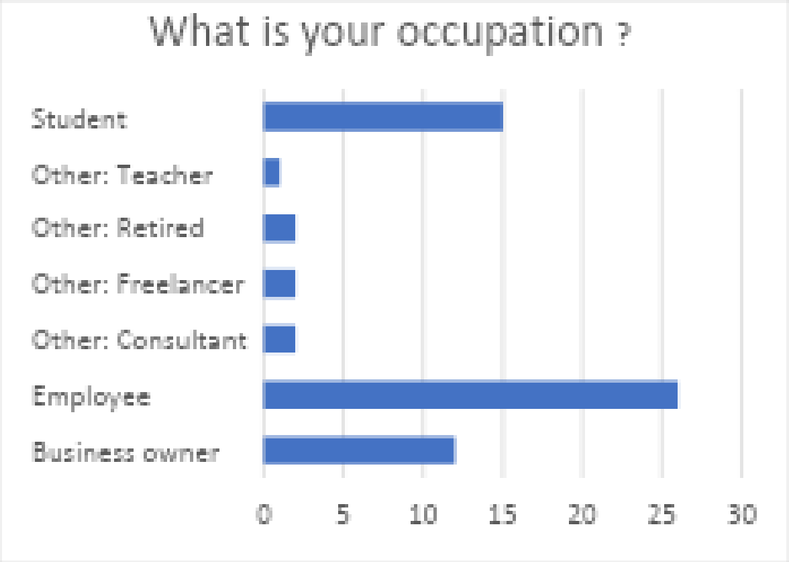


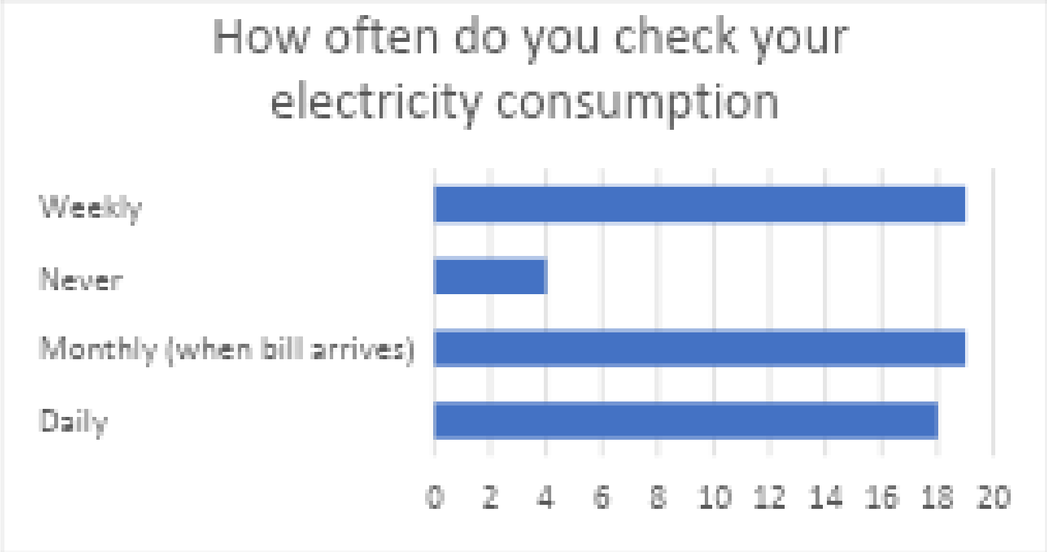
general idea

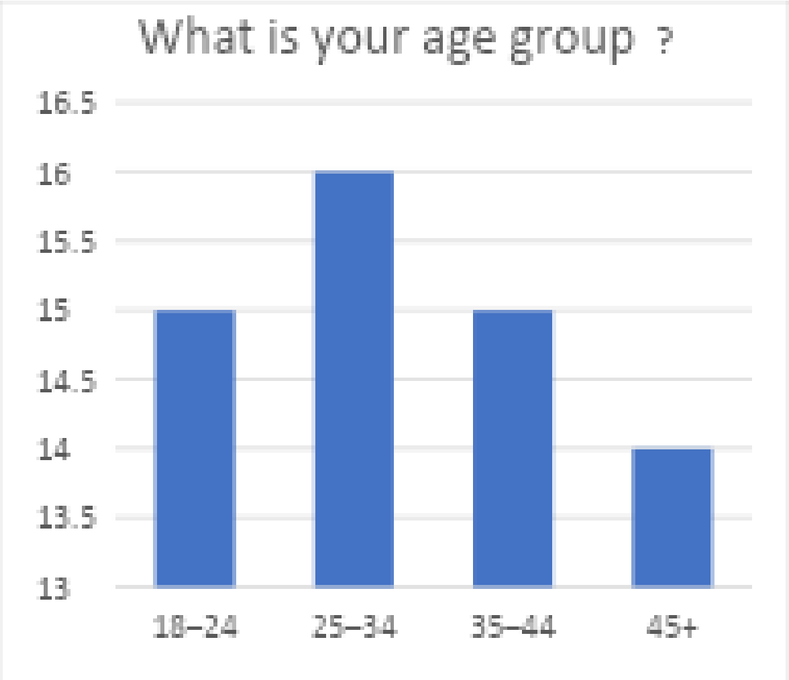
Our Smart Meter Mobile Application will let users monitor their electricity consumption in real time, receive intelligent alerts, and access personalized energy-saving recommendations, which will affect homeowners like Ahmed who currently have no reliable way to track or manage their energy usage throughout the month,

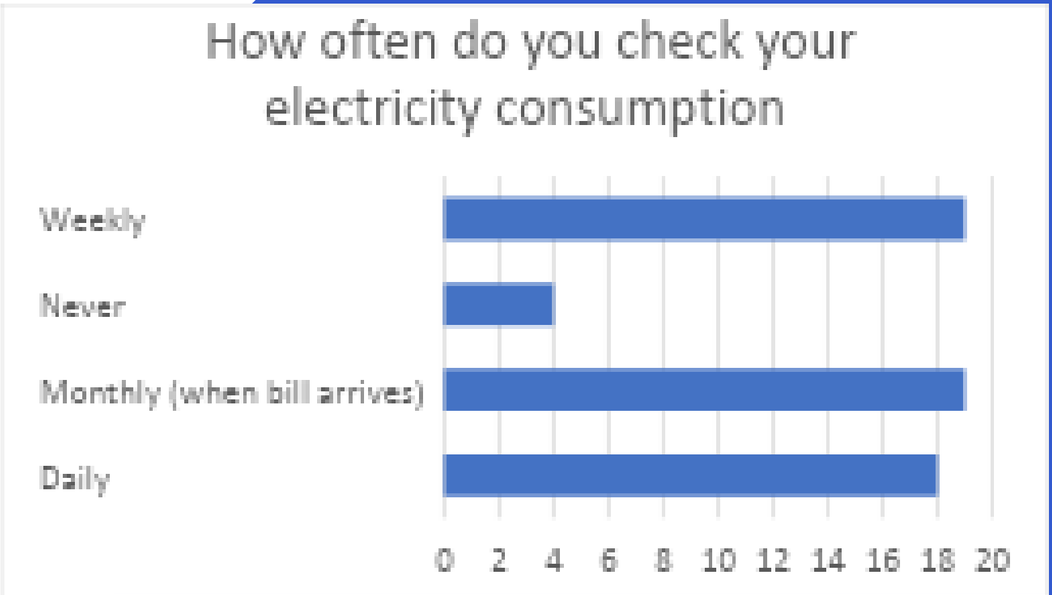
by reducing bill shock, eliminating energy waste, and building confidence in managing household electricity efficiently. We will measure effectiveness by tracking reductions in average monthly electricity consumption, user-reported bill savings, frequency of alert engagement, and overall user satisfaction scores.

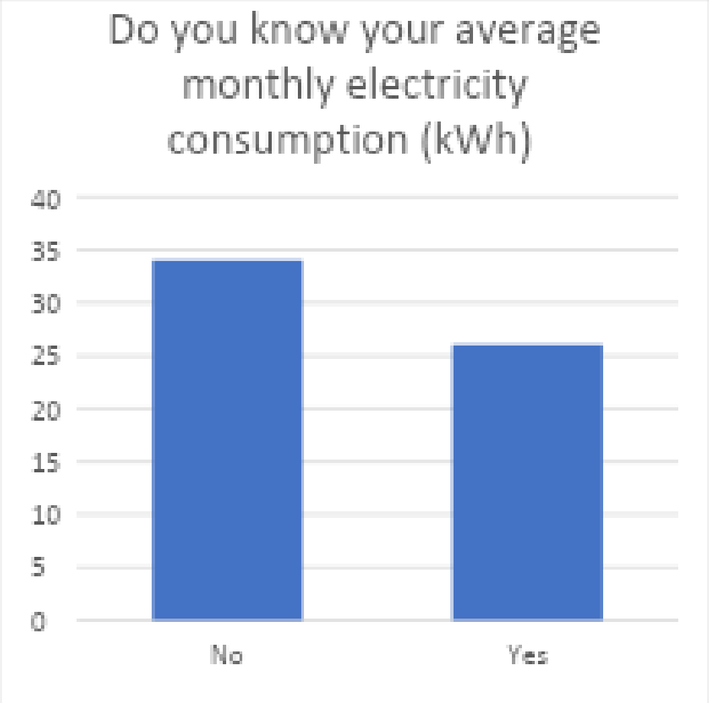
Research

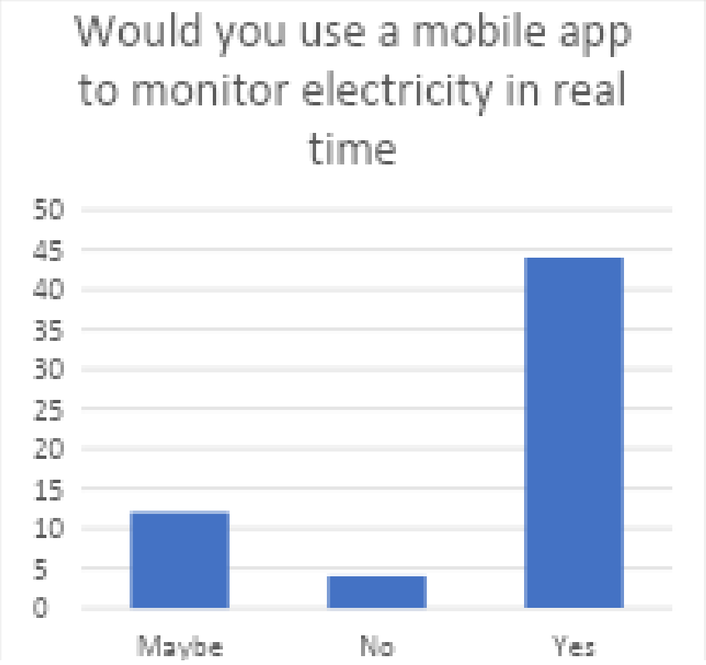

What is your occupation ?
Student
Other: Teacher
Other: Retired
Other: Freelancer
Other: Consultant
Employee
Business owner
0
5
10
15
20
25
30

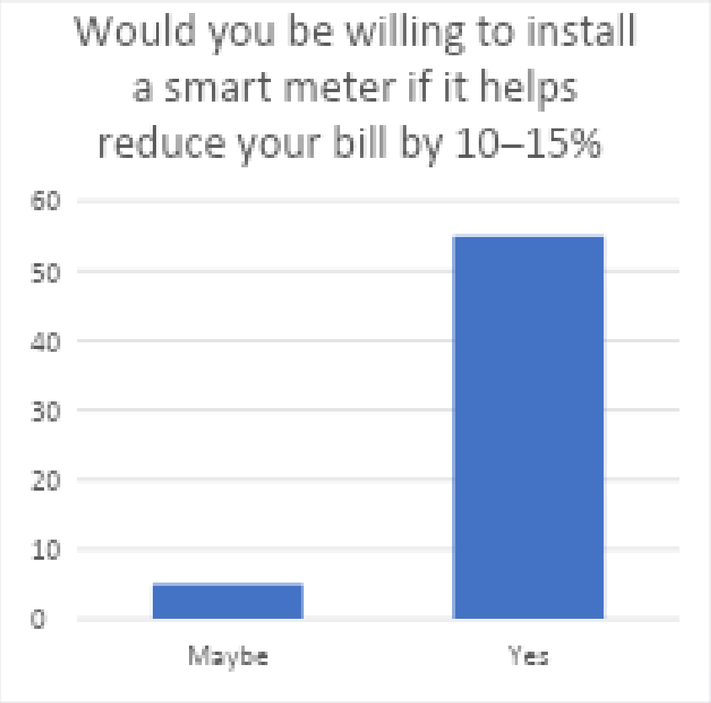

How often do you check your electricity consumption
Weekly
Never
Monthly (when bill arrives)
Daily
0
2
4
6
8
10
12
14
16
18
20

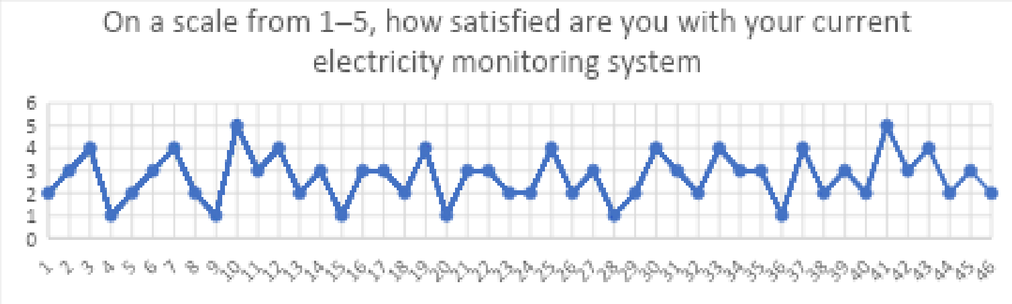

What is your age group ?
16.5
16
15.5
15
14.5
14
13.5
13
18–24
25–34
35–44
45+

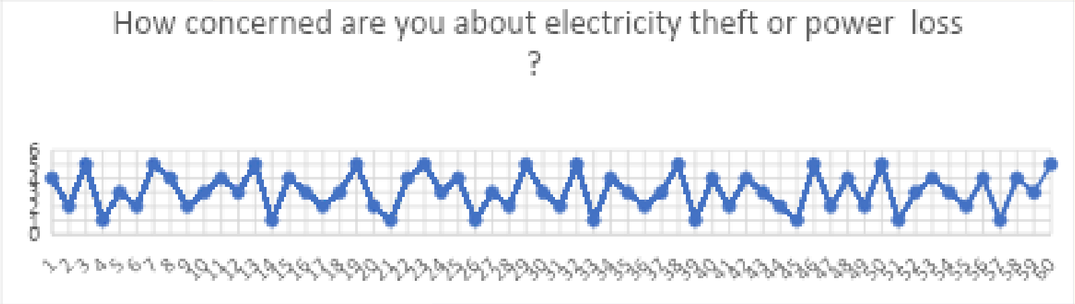

How often do you check your electricity consumption
Weekly
Never
Monthly (when bill arrives)
Daily
0
2
4
6
8
10
12
14
16
18
20

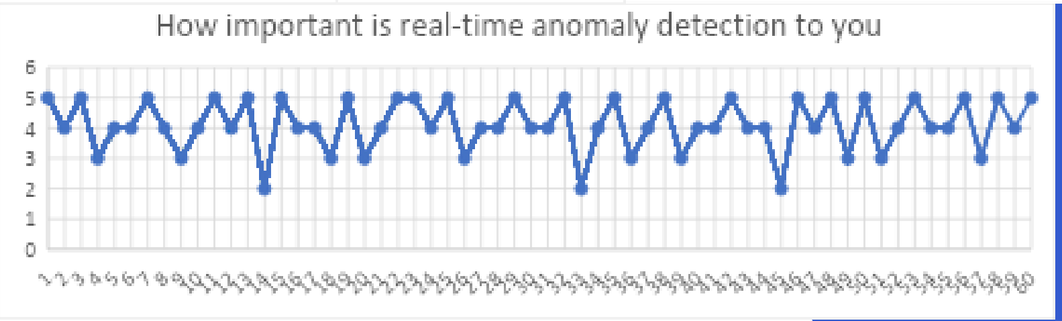

Do you know your average monthly electricity consumption (kWh)
40
35
30
25
20
15
10
5
0
No
Yes


Would you use a mobile app to monitor electricity in real time
50
45
40
35
30
25
20
15
10
5
0
Maybe
No
Yes


Would you be willing to install a smart meter if it helps reduce your bill by 10–15%
60
50
40
30
20
10
0
Maybe
Yes


On a scale from 1–5, how satisfied are you with your current electricity monitoring system
6
5
4
3
2
1
0


How concerned are you about electricity theft or power loss ?


How important is real-time anomaly detection to you
6
5
4
3
2
1
0

Persona

The following persona was developed through user research to represent the primary target user of the Smart Meter Mobile Application. Ahmed's profile informs all subsequent design and problem-definition decisions.

Ahmed Hassan

Age : 35

Education : Masters

Occupation : Homeowner

Location : Cairo , Egypt

Status : Marriad

Bio

Ahmed is a tech-savvy homeowner who wants better control and visibility over his household energy consumption. He is concerned about energy waste, cost, and ensuring his family's comfort.

Interests

Smart home technology, energy efficiency, IoT devices, renewable energy

Goals

- Monitor household energy consumption in real-time
- Reduce electricity bills
- Ensure devices operate efficiently
- Detect unusual consumption

Motivation

- Save money on energy bills
- Protect his home and appliances
- Contribute to sustainable energy use

Needs & Expectations

- Reliable, real-time monitoring
- Alerts for unusual usage
- Secure and easy-to-use interface (mobile & web)
- Integration with other smart home devices

Challenges

- Understanding complex energy data
- Dealing with unreliable or insecure IoT devices
- Limited control over energy distribution in older homes

Pain Points

- Unreliable smart meters
- Lack of transparency in energy consumption
- Fear of cyber threats on connected devices
- Difficulty detecting appliance faults

Empathy Map: Ahmed Hassan

Says

- I never know my electricity consumption until the bill arrives
- Sometimes my bill is much higher than I expect
- I wish I could see which devices use the most electricity
- Current electricity monitoring systems are confusing or hard to use
- It's hard to track my energy usage on a daily basis
- I want alerts if something abnormal happens with my electricity
- I don't understand my bill calculations
- I wish there was an easy way to save money on electricity

Thinks

- Am I wasting electricity without realizing it?
- Maybe my air conditioner or heater is consuming too much
- Is there a way to track my usage in real time?
- A smart meter might help me reduce my monthly bills
- I want to identify faulty or inefficient devices in my home
- Can I prevent sudden high bills?
- Maybe technology can help me manage energy more efficiently
- I want control over my electricity usage without too much effort

Empathy Map: Ahmed Hassan

Does

- Checks electricity bill only when it arrives
- Tries to save electricity by turning off devices randomly
- Keeps track of appliances manually, if at all
- Asks family or friends for tips to reduce electricity consumption
- Sometimes switches off devices to save power without knowing real impact
- Uses basic energy-saving habits (like unplugging chargers)
- Monitors usage occasionally through home appliances' indicators
- Keeps notes or spreadsheets to track high consumption periods

Feels

- Anxious about unexpectedly high bills
- Frustrated by lack of control over electricity usage
- Hopeful that a smart meter could improve monitoring
- Overwhelmed by confusing bills
- Worried about energy waste
- Confused about which appliance consumes the most
- Relieved when energy usage is predictable
- Motivated to save money with better insights

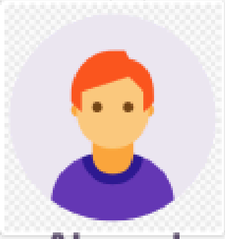
Empathy Mapping

Says

- I never know my electricity consumption until the bill arrives
- Sometimes my bill is much higher than I expect
- I wish I could see which devices use the most electricity.
- Current electricity monitoring systems are confusing or hard to use.
- It's hard to track my energy usage on a daily basis
- I want alerts if something abnormal happens with my electricity
- I don't understand my bill calculations.
- I wish there was an easy way to save money on electricity

Thinks

- Am I wasting electricity without realizing it?
- Maybe my air conditioner or heater is consuming too much
- Is there a way to track my usage in real time?
- A smart meter might help me reduce my monthly bills.
- I want to identify faulty or inefficient devices in my home
- Can I prevent sudden high bills?
- Maybe technology can help me manage energy more efficiently
- "I want control over my electricity usage without too much effort."



Ahmed

Does

- Checks electricity bill only when it arrives.
- Tries to save electricity by turning off devices randomly.
- Keeps track of appliances manually, if at all.
- Asks family or friends for tips to reduce electricity consumption.
- Sometimes switches off devices to save power without knowing real impact.
- Uses basic energy-saving habits (like unplugging chargers).
- Monitors usage occasionally through home appliances' indicators.
- Keeps notes or spreadsheets to track high consumption periods.

Feels

- Anxious about unexpectedly high bills.
- Frustrated by lack of control over electricity usage.
- Hopeful that a smart meter could improve monitoring.
- Overwhelmed by confusing bills.
- Worried about energy waste.
- Confused about which appliance consumes the most.
- Relieved when energy usage is predictable.
- Motivated to save money with better insights.

Pain Points & Categories

Category	Description
Financial	Money-related issues — Ahmed receives unexpectedly high electricity bills with no clear breakdown of what drove the cost.
Product	Quality issues — Smart meters and monitoring apps are unreliable, inaccurate, or difficult to set up and use.
Process	Related to the user's journey — Ahmed has no consistent routine to track usage; he relies on random manual checks.
Support	Getting help from customer service — Limited guidance from utility providers on how to interpret bills or reduce usage.

Ahmed's Key Pain Points
● Overwhelming lack of real-time visibility into daily electricity consumption
● Unexpectedly high bills with no clear explanation of what caused the spike
● Unreliable or confusing existing monitoring systems and smart meters
● No automatic alerts when consumption is unusually high or abnormal
● Limited understanding of which appliances are the biggest energy consumers
● Fear of cyber threats and data privacy concerns with connected IoT devices

3.2 5Ws and H Analysis

To create an accurate problem statement, the 5Ws and H framework was applied using insights drawn from the persona and empathy map. This analysis grounds the problem in Ahmed's real-world context:

Question	Answer
Who?	Ahmed Hassan — a 35-year-old homeowner in Cairo who wants real-time visibility into his household electricity consumption.
What?	Ahmed struggles to monitor and control his electricity usage in real time. He has no reliable tool to track which devices consume the most or to receive alerts when usage is unusually high.
Where?	At home — primarily through a mobile app while managing daily household routines.
When?	The problem occurs continuously throughout the month, with peak frustration when the electricity bill arrives unexpectedly high.
Why?	Lack of real-time data causes Ahmed to waste energy unknowingly, leading to financial strain and anxiety about the bill.
How?	Ahmed currently checks his bill only on arrival, turns off devices randomly, and manually estimates usage — all without any data-driven insight.

Ideate a Solution

Stage 03 marks the transition from the problem space to the solution space. Having defined Ahmed's pain points and problem statement, this stage develops a Goal Statement that sets the scope and direction for the Smart Meter Mobile Application design.

4.1 Goal Statement Approach

To focus the scope of our design, we create a Goal Statement: one or two sentences that describe the product and its benefits for the user. A well-formed goal statement answers three key questions:

Who	Who will the product serve? — Ahmed: a tech-savvy homeowner seeking real-time energy visibility.
What	What will the product do? — Provide real-time monitoring, usage dashboards, intelligent alerts, and energy-saving recommendations.
Why	Why does the product solve the user's need? — It eliminates the feedback delay between usage and awareness, empowering Ahmed to make informed, timely decisions about his energy consumption.

Goal Statement

Our Smart Meter Mobile Application **will let users** monitor their electricity consumption in real time, receive intelligent alerts, and access personalized energy-saving recommendations, **which will affect** homeowners like Ahmed who currently have no reliable way to track or manage their energy usage throughout the month, **by** reducing bill shock, eliminating energy waste, and building confidence in managing household electricity efficiently. **We will measure effectiveness by** tracking reductions in average monthly electricity consumption, user-reported bill savings, frequency of alert engagement, and overall user satisfaction scores.

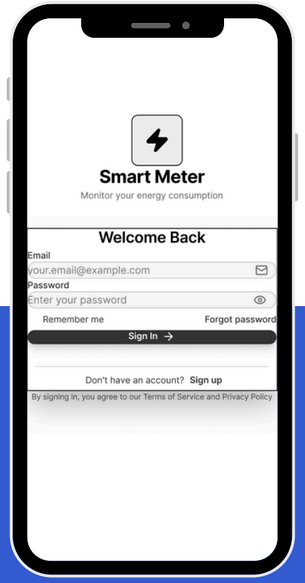
4.3 Proposed Application Features

1. directly from the smart meter, giving Ahmed instant visibility into current household usage.
2. Usage Dashboard — Visualizes daily, weekly, and monthly consumption trends through intuitive charts, enabling Ahmed to spot patterns and compare periods at a glance.
3. Alerts & Notifications — Automatically notifies Ahmed when consumption exceeds a threshold or when an unusual pattern is detected — addressing his fear of surprise bills.
4. Energy Saving Recommendations — Delivers personalized, data-driven suggestions to help Ahmed reduce usage and identify inefficient appliances.
5. Usage History — Provides a complete historical log of consumption data, allowing Ahmed to track progress, review past periods, and understand his long-term energy habits.

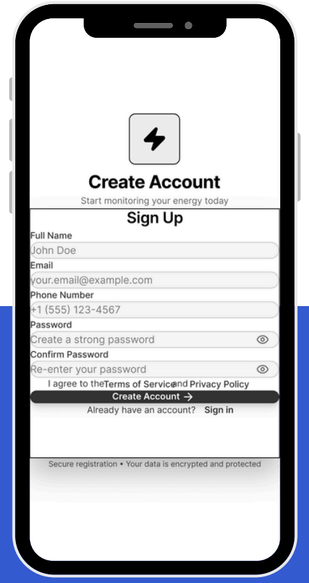
Problem Statement

Ahmed is **a/an** tech-savvy homeowner who manages household energy manually **who needs** a reliable, real-time energy monitoring mobile application **because** the lack of visibility into his electricity consumption leads to unexpectedly high bills, energy waste, and anxiety — with no easy way to identify inefficient appliances or receive timely alerts about abnormal usage.

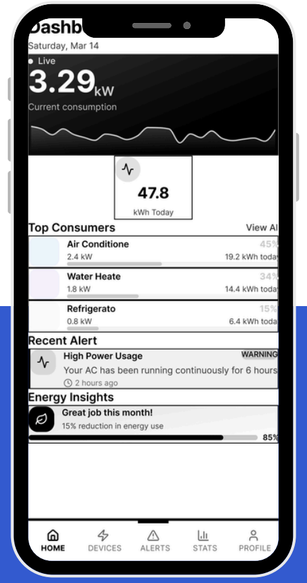
UI / UX



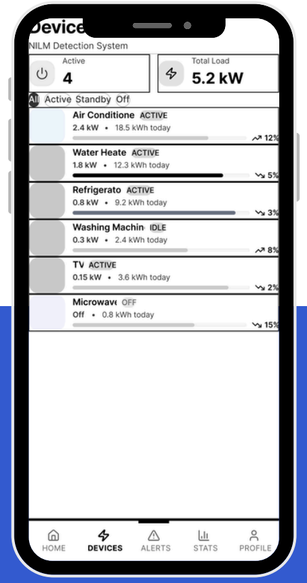
Sign-in



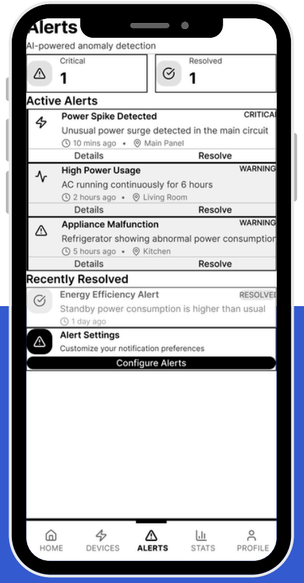
Sign-up



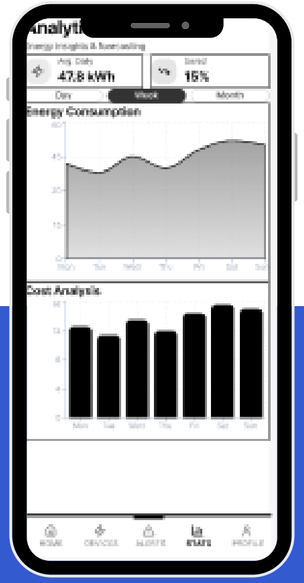
Home



Devices



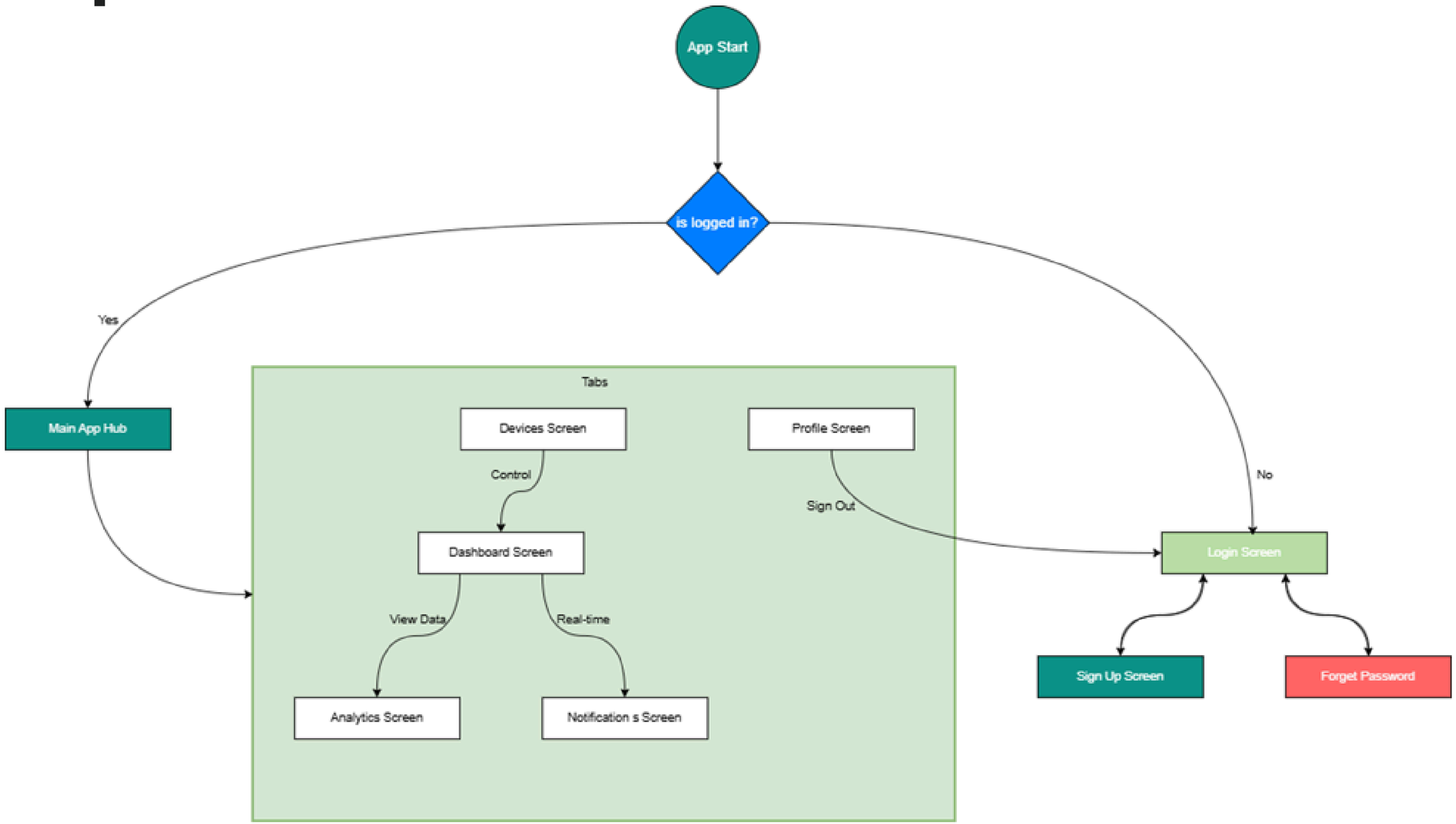
Alarts



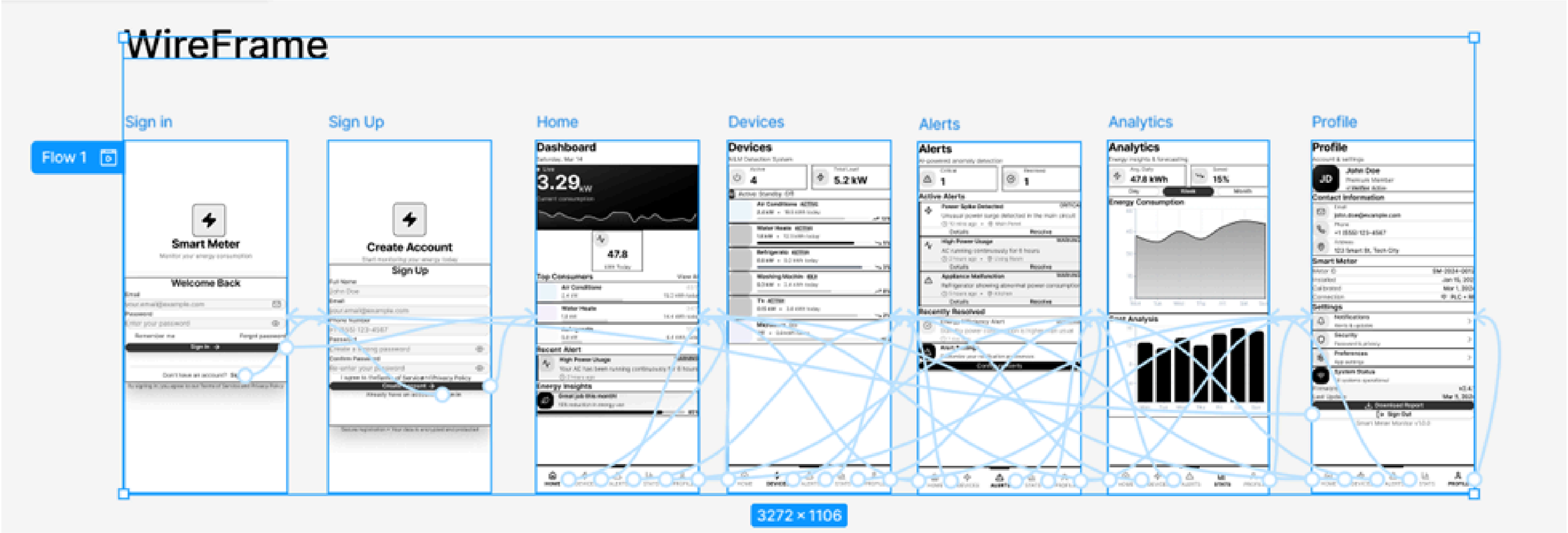
Analytics

Profile

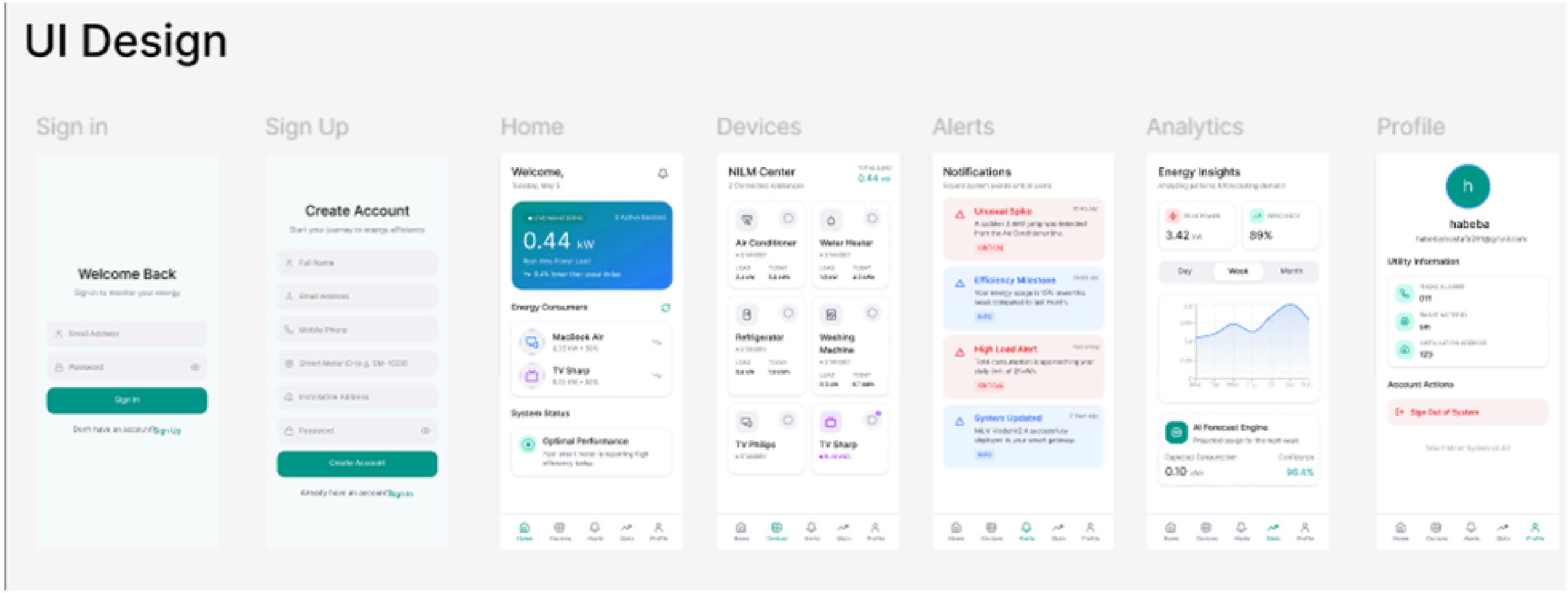
Site Map


App Start
is logged in?
Yes
No
Main App Hub
Tabs
Devices Screen
Profile Screen
Control
Sign Out
Dashboard Screen
View Data
Real-time
Analytics Screen
Notification s Screen
Login Screen
Sign Up Screen
Forget Password

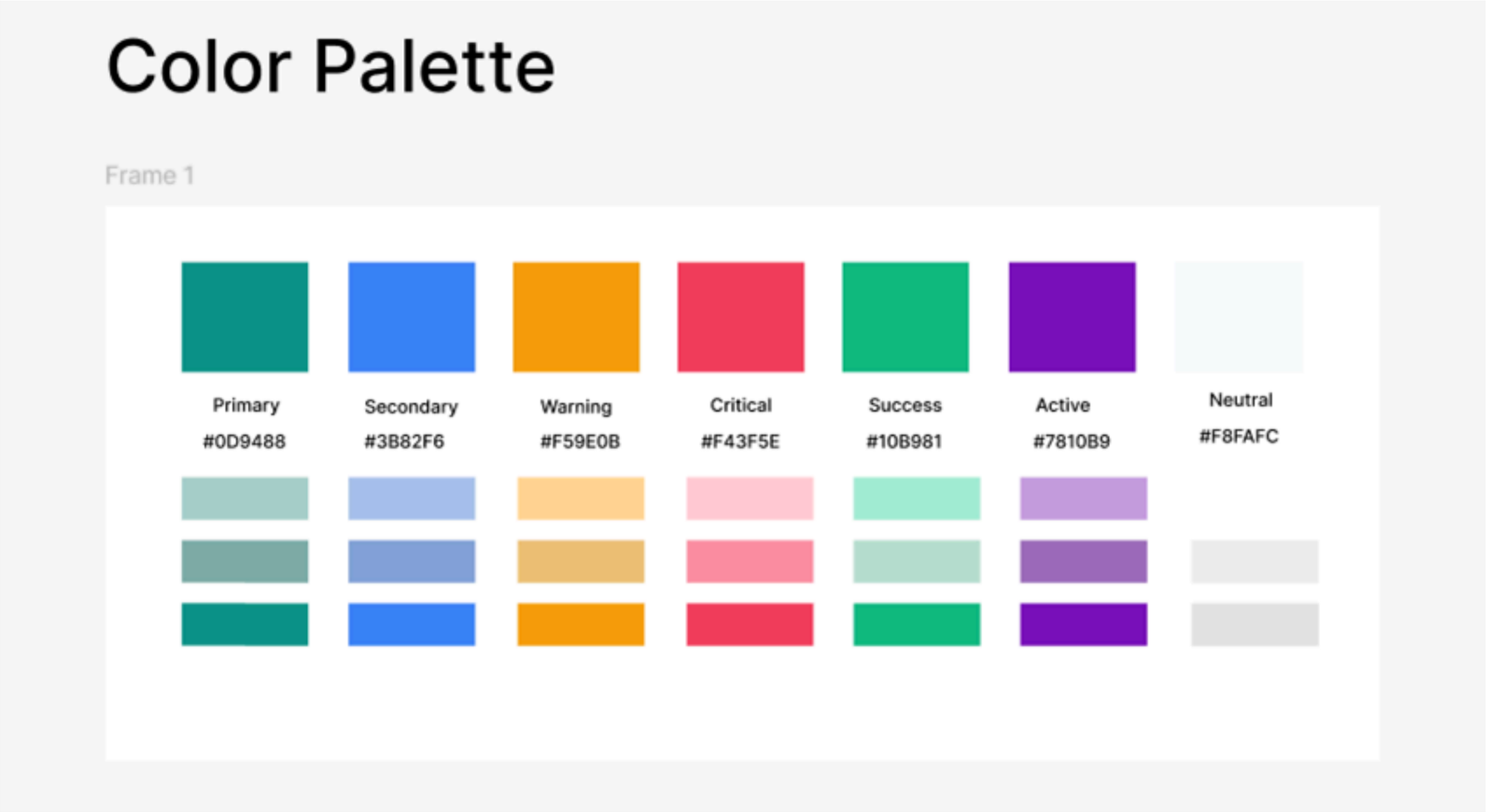
Wirframe and it is interactions



UI Design


UI Design
Sign in
Welcome Back
Sign Up
Create Account
Home
Welcome,
0.44 kW
Energy Consumers
MacBook Air
TV Sharp
System Status
Optimal Performance
Devices
NILM Center
Air Conditioner
Water Heater
Refrigerator
Washing Machine
TV Philips
TV Sharp
Alerts
Notifications
Unusual Spike
Efficiency Milestone
High Load Alert
System Updated
Analytics
Energy Insights
3.42
89%
Day
Week
Month
AI Forecast Engine
0.10
96.4%
Profile
habeba
Utility Information
Account Actions
Sign Out of System

Frontend


Color Palette
Frame 1
Primary
#0D9488
Secondary
#3B82F6
Warning
#F59E0B
Critical
#F43F5E
Success
#10B981
Active
#7810B9
Neutral
#F8FAFC

Typography (Font Hierarchy)

Role	Font Name	Size	Usage
Title	Inter Bold	32 - 48px	App name, Main headers, Large power values
Header	Inter SemiBold	20 - 24px	Screen titles, Section headers, Card titles
Regular	Inter Regular	14 - 16px	Body text, descriptions, form labels
Bold	Inter Bold	14 - 18px	Button labels, high-load values, statistics

Custom Buttons

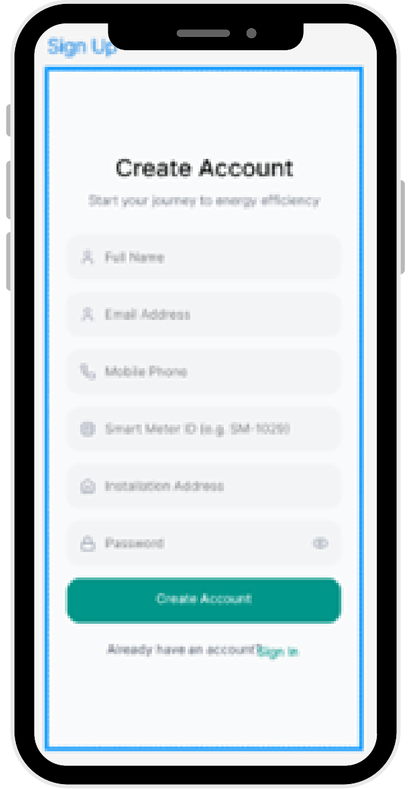
Component Name	Style	Where is used?	Component Design Brief
PrimaryButton	Solid Teal, Rounded	Login, Sign Up, Save	High-contrast, 16px radius, full-
SecondaryButton	Outlined Indigo	Logout, Cancel, Reset	Transparent background, 2px
AuthToggle	Text Link	Login <-> SignUp	Subtle typography link with Primary

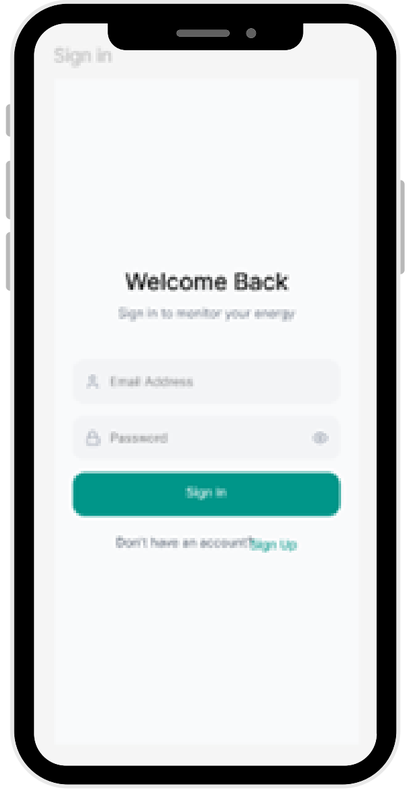
Other Custom Components

Component Name	Style	Where is used?	Component Design Brief
EnergyCard	Glassmorphism	Dashboard, Devices	Semi-transparent background with
AuthForm	Unified Container	Auth Flow	Single component that switches
AlertBanner	Floating Toast	Dashboard (Top)	Critical rose background, slides
PulseIcon	Animated SVG	Dashboard	Teal circle with a "breathing"

Screenshots for UI Design

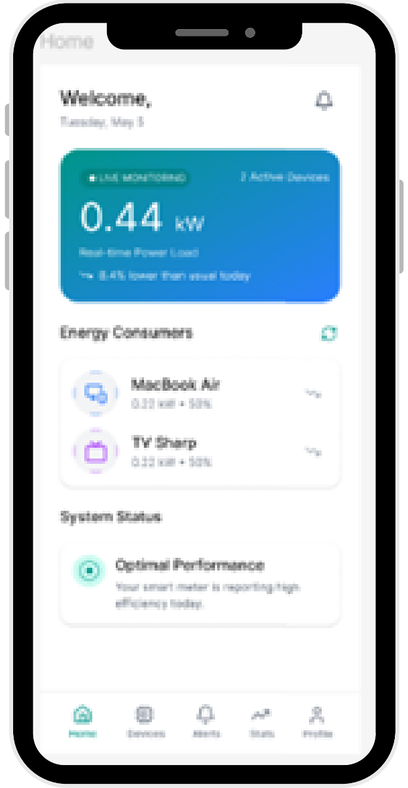
1. The Unified Auth Gateway (Login/Sign-Up)
Description: A minimalist, high-contrast entry point that uses a unified AuthForm component. It features a clean white background with Energy Teal accents. The design focuses on "Zero-Distraction" UX, using smooth transitions between Login and Registration modes to reduce user friction.





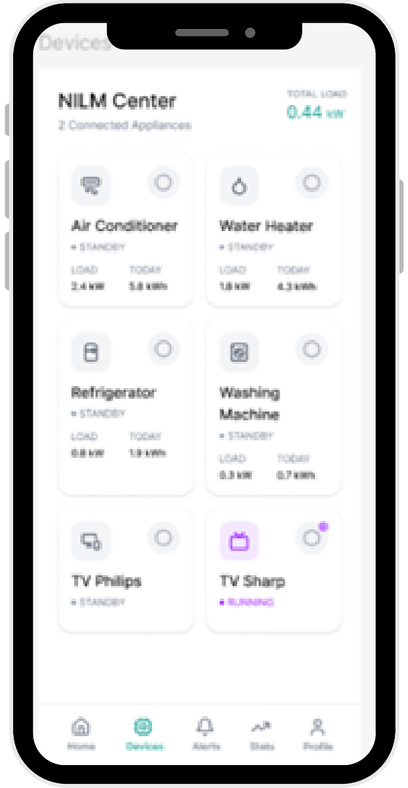
Screenshots for UI Design

2. The Live Intelligence Dashboard
Description: The heart of the application. It utilizes a Glassmorphic Hero Card that displays a real-time "Consumption Wave." A breathing pulse animation indicates active AI monitoring. The layout uses a 2-column grid for quick stats (Daily Cost, Voltage, Current), ensuring the most critical data is visible at a glance.



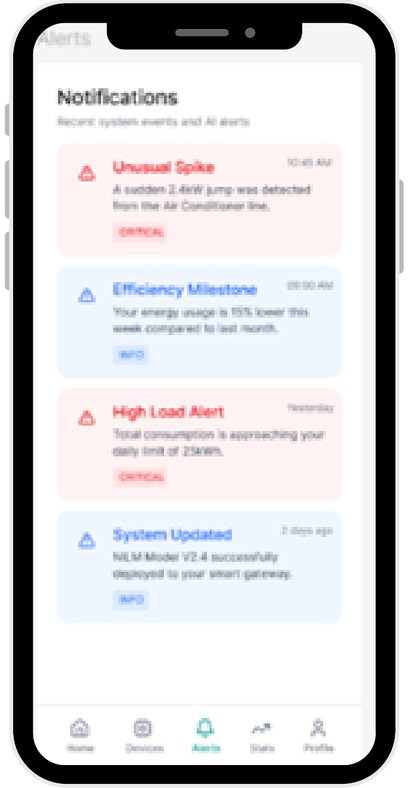
Screenshots for UI Design

3. The Appliance Control Hub (Devices)
Description: A vibrant, card-based interface showing individual appliance statuses. Each card features a Staggered Entrance Animation and a "Glow State"—a soft color aura (e.g., Emerald for Fridge, Amber for AC) that appears only when the AI detects the device is active.



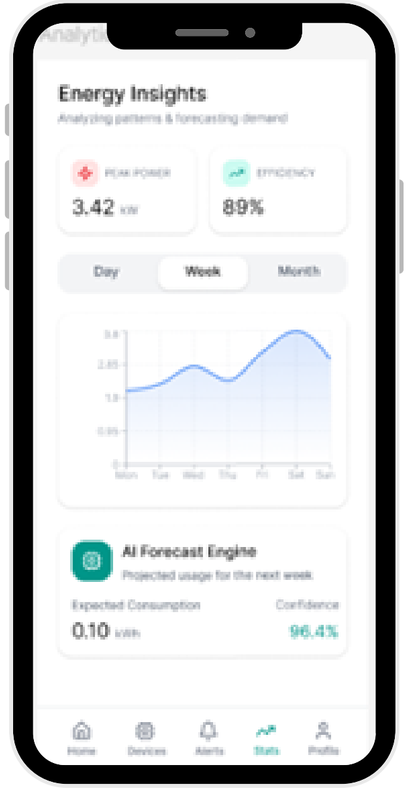
Screenshots for UI Design

4. The Deep-Insight Analytics
Description: A sophisticated Dark-Mode analytical suite. It uses a deep Indigo-to-Black gradient to make the energy forecasting charts "pop." The screen includes an AI Confidence badge and a dynamic wave chart that visualizes predicted consumption patterns for the upcoming week.



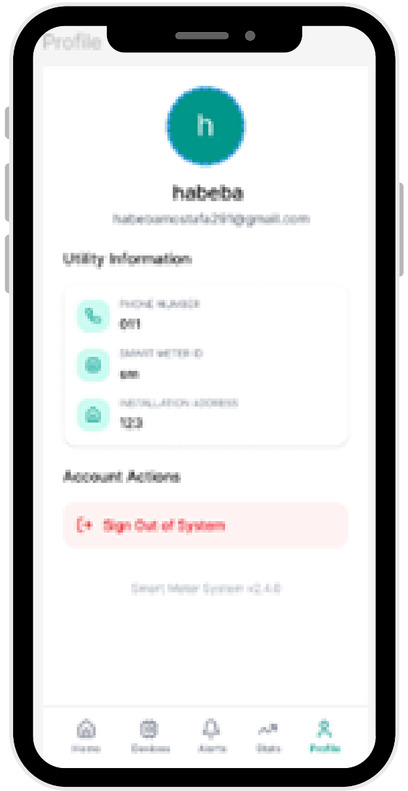
Screenshots for UI Design

5. The Smart Notification Center (Alerts)
Description: An elegant, button-free notification log. Alerts are categorized by color: Critical Rose for energy spikes and Data Indigo for efficiency updates. The design uses semi-transparent "Glass" rows with 24px corner radii, following the modern mobile "Toast" notification aesthetic.

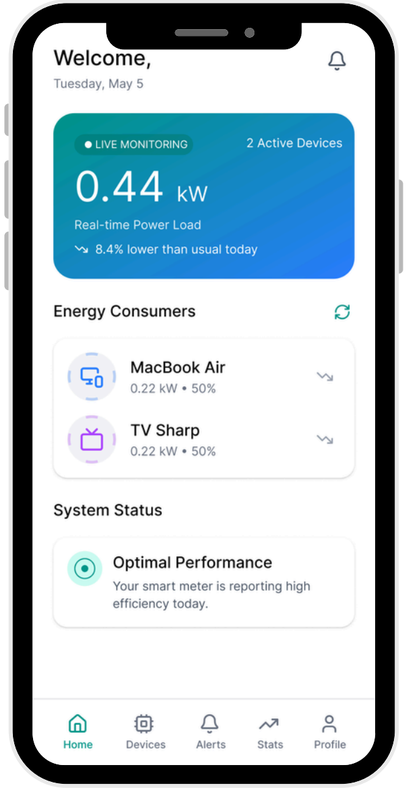


Screenshots for UI Design

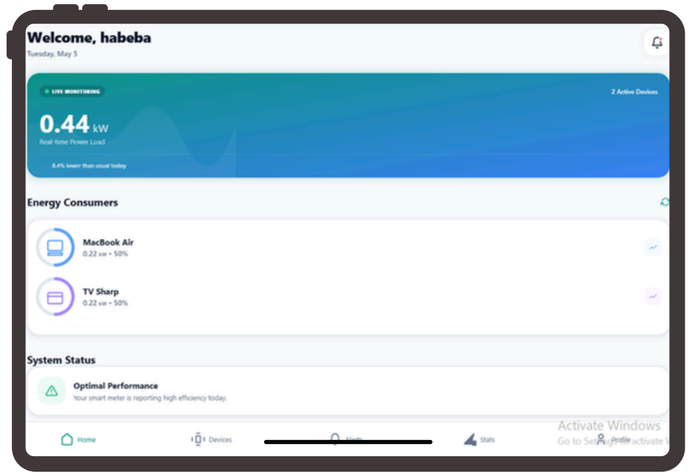
6. The Utility Command Center (Profile)
Description: A personalized space for the user. It features a Glowing Avatar Ring and a structured "Information Grid" that fetches the User's Phone, Meter ID, and Home Address from Firestore.



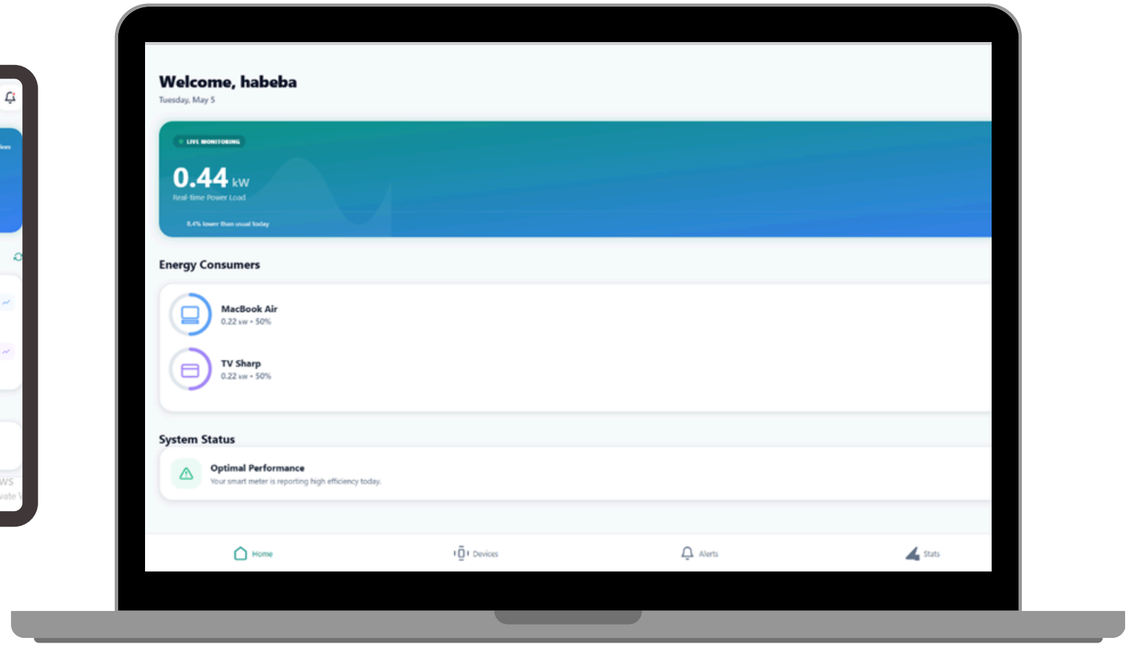
Responsiveness



Mobile View



Tablet View



Desktop View

Backend & Dynamic Architecture

1. Core Infrastructure (Firebase & Firestore)
The backbone of the Smart Meter Monitoring System is built on Google Firebase. We utilized a NoSQL Cloud Firestore database to handle real-time data streaming. Unlike traditional databases, Firestore allows the app to listen for changes and update the UI instantly without the user needing to refresh the page.

2. The Dynamic Content Pipeline
Your app’s content is not static; it "breathes" with the incoming energy data. Here is the technical flow of how data is rendered:
Data Ingestion: A Python script or a physical IoT gateway pushes raw electrical parameters (Voltage, Current, Active Power) into the Firestore data_nilm collection every few seconds.

State Synchronization: Inside the React Native app, we use React Hooks (useEffect and useState) combined with Firestore queries to fetch the most recent 400 entries of data.

AI Processing (The Intelligence Layer):
Once the app retrieves the dynamic data from the database, it doesn't just show numbers.
It pipes this raw data into our Hugging Face AI Models.
The backend processes this via an API call and returns a "Disaggregated Load" (NILM) and "Anomaly Score."

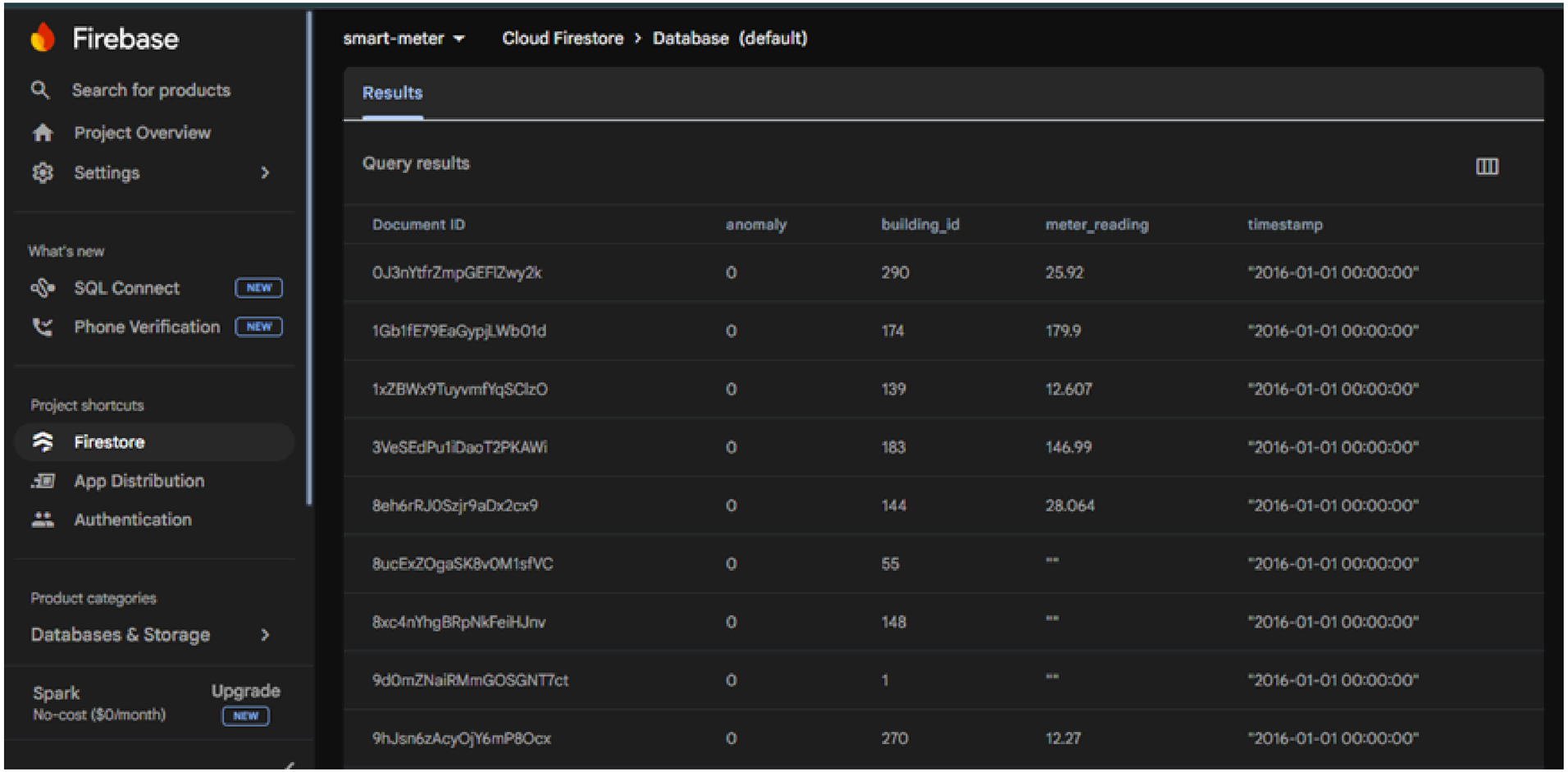
Backend Screens

Backend Technology Stack Table

Technology	Role	Description
Cloud Firestore	Primary Database	Real-time NoSQL storage for energy logs and user metadata.
Firebase Auth	Security Layer	Handles encrypted Login, Sign-Up, and session management.
Hugging Face API	AI Inference	Hosts our NILM, Anomaly, and Forecast models on a FastAPI server.
Python (Backend)	Data Simulation	Used for cleaning and uploading raw CSV datasets into the live cloud stream.

Backend Screens

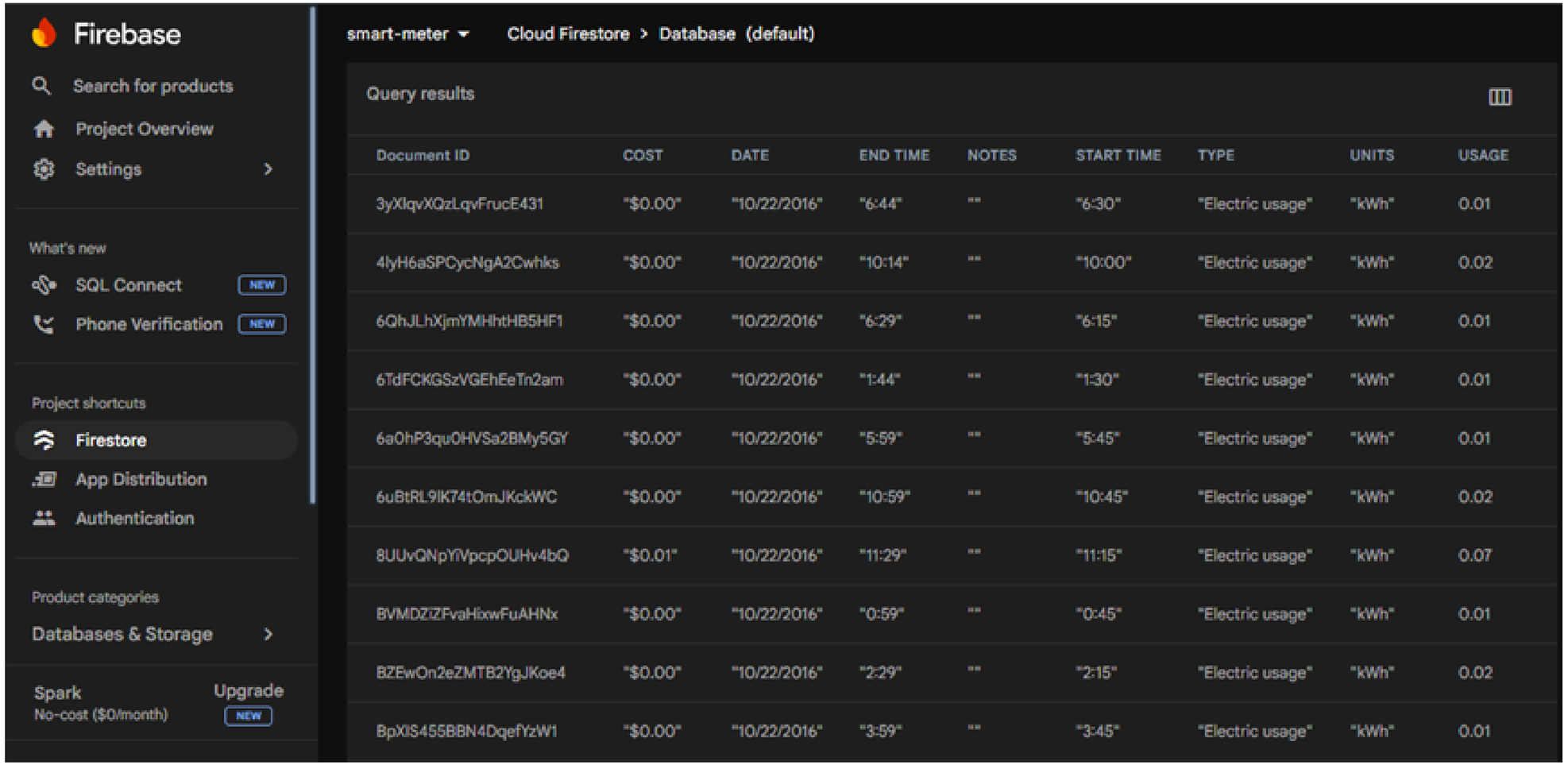
data_anomaly



Document ID	anomaly	building_id	meter_reading	timestamp
0J3nYtfrZmpGEFlZwy2k	0	290	25.92	"2016-01-01 00:00:00"
1Gb1fE79EaGypjLWb01d	0	174	179.9	"2016-01-01 00:00:00"
1xZBWx9TuyvmfYqSClzO	0	139	12.607	"2016-01-01 00:00:00"
3VeSEdPu1iDaoT2PKAWi	0	183	146.99	"2016-01-01 00:00:00"
8eh6rRJ0Szjr9aDx2cx9	0	144	28.064	"2016-01-01 00:00:00"
8ucExZOgaSK8v0M1sfVC	0	55	""	"2016-01-01 00:00:00"
8xc4nYhgBRpNkFeiHJnv	0	148	""	"2016-01-01 00:00:00"
9d0mZNaiRMmGOSGNT7ct	0	1	""	"2016-01-01 00:00:00"
9hJsn6zAcyOjY6mP8Ocx	0	270	12.27	"2016-01-01 00:00:00"

Backend Screens

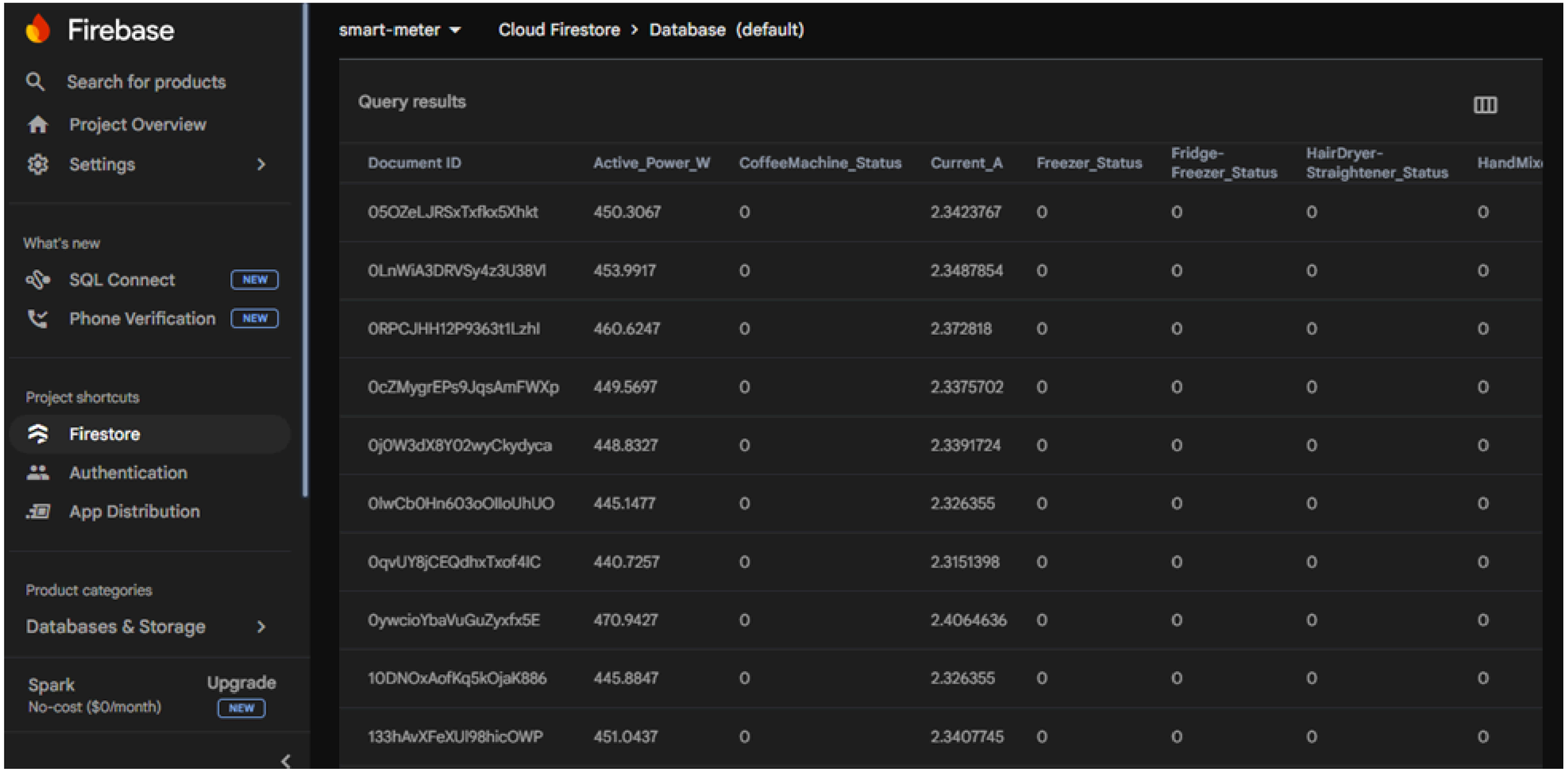
data_forecast



Document ID	COST	DATE	END TIME	NOTES	START TIME	TYPE	UNITS	USAGE
3yXlqvXQzLqvFrucE431	"$0.00"	"10/22/2016"	"6:44"	""	"6:30"	"Electric usage"	"kWh"	0.01
4lyH6aSPCycNgA2Cwhks	"$0.00"	"10/22/2016"	"10:14"	""	"10:00"	"Electric usage"	"kWh"	0.02
6QhJLhXjmYMHhtHB5HF1	"$0.00"	"10/22/2016"	"6:29"	""	"6:15"	"Electric usage"	"kWh"	0.01
6TdFCKGSzVGEhEeTn2am	"$0.00"	"10/22/2016"	"1:44"	""	"1:30"	"Electric usage"	"kWh"	0.01
6aOhP3qu0HVSa2BMy5GY	"$0.00"	"10/22/2016"	"5:59"	""	"5:45"	"Electric usage"	"kWh"	0.01
6uBtRL9IK74tOmJKckWC	"$0.00"	"10/22/2016"	"10:59"	""	"10:45"	"Electric usage"	"kWh"	0.02
8UUvQNpYiVpcpOUHv4bQ	"$0.01"	"10/22/2016"	"11:29"	""	"11:15"	"Electric usage"	"kWh"	0.07
BVMDZiZFvaHixwFuAHNx	"$0.00"	"10/22/2016"	"0:59"	""	"0:45"	"Electric usage"	"kWh"	0.01
BZEwOn2eZMTB2YgJKoe4	"$0.00"	"10/22/2016"	"2:29"	""	"2:15"	"Electric usage"	"kWh"	0.02
BpXIS455BBN4DqefYzW1	"$0.00"	"10/22/2016"	"3:59"	""	"3:45"	"Electric usage"	"kWh"	0.01

Backend Screens

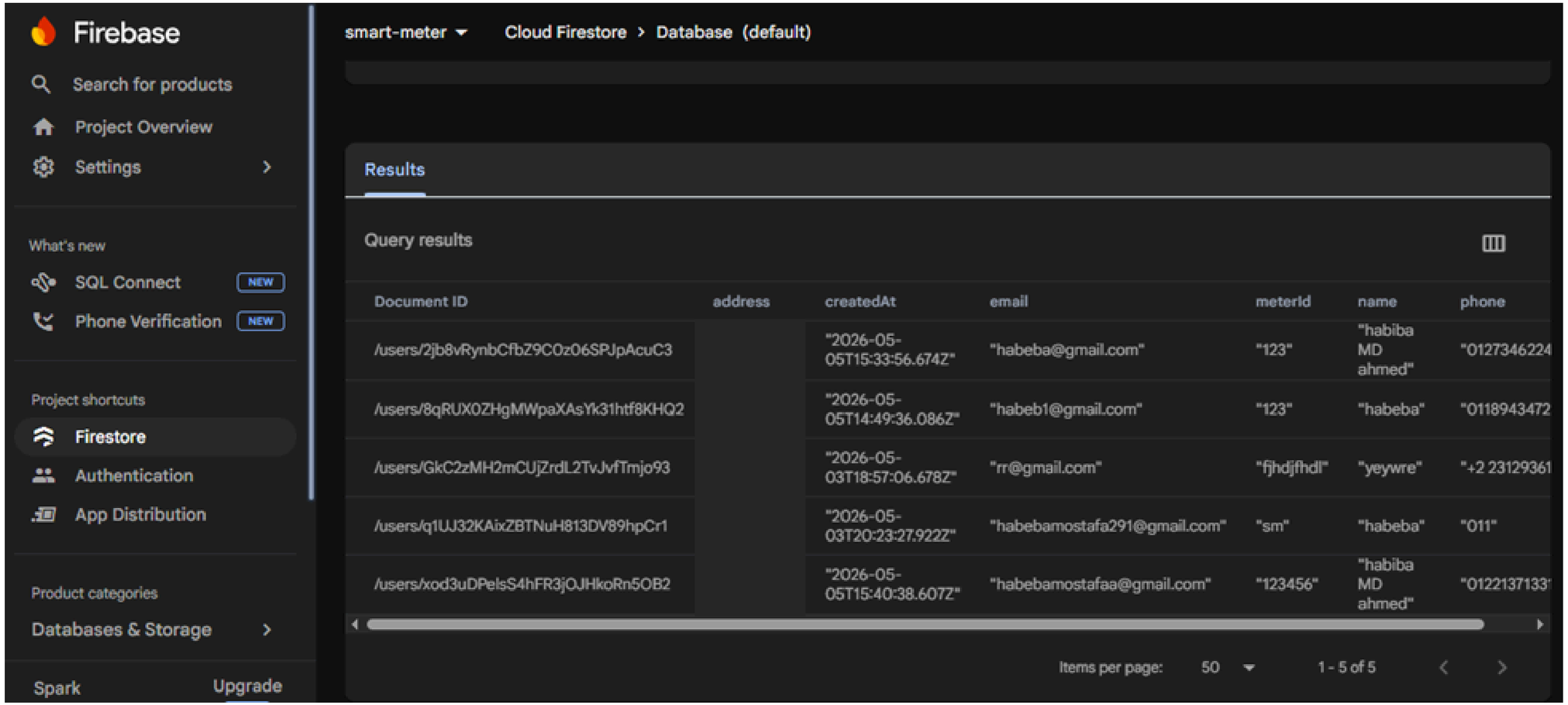
data_nilm



Document ID	Active_Power_W	CoffeeMachine_Status	Current_A	Freezer_Status	Fridge-Freezer_Status	HairDryer-Straightener_Status	HandMix
05OZeLJRSxTxflox5Xhkt	450.3067	0	2.3423767	0	0	0	0
0LnWiA3DRVSy4z3U38Vl	453.9917	0	2.3487854	0	0	0	0
0RPCJHH12P9363t1LzhI	460.6247	0	2.372818	0	0	0	0
0cZMygrEPs9JqsAmFWXp	449.5697	0	2.3375702	0	0	0	0
0j0W3dX8Y02wyCkydyca	448.8327	0	2.3391724	0	0	0	0
0lwCb0Hn6O3oOIloUhUO	445.1477	0	2.326355	0	0	0	0
0qvUY8jCEQdhxTxof4IC	440.7257	0	2.3151398	0	0	0	0
0ywcioYbaVuGuZyxfx5E	470.9427	0	2.4064636	0	0	0	0
10DNOxAofKq5kOjaK886	445.8847	0	2.326355	0	0	0	0
133hAvXFeXUI98hicOWP	451.0437	0	2.3407745	0	0	0	0

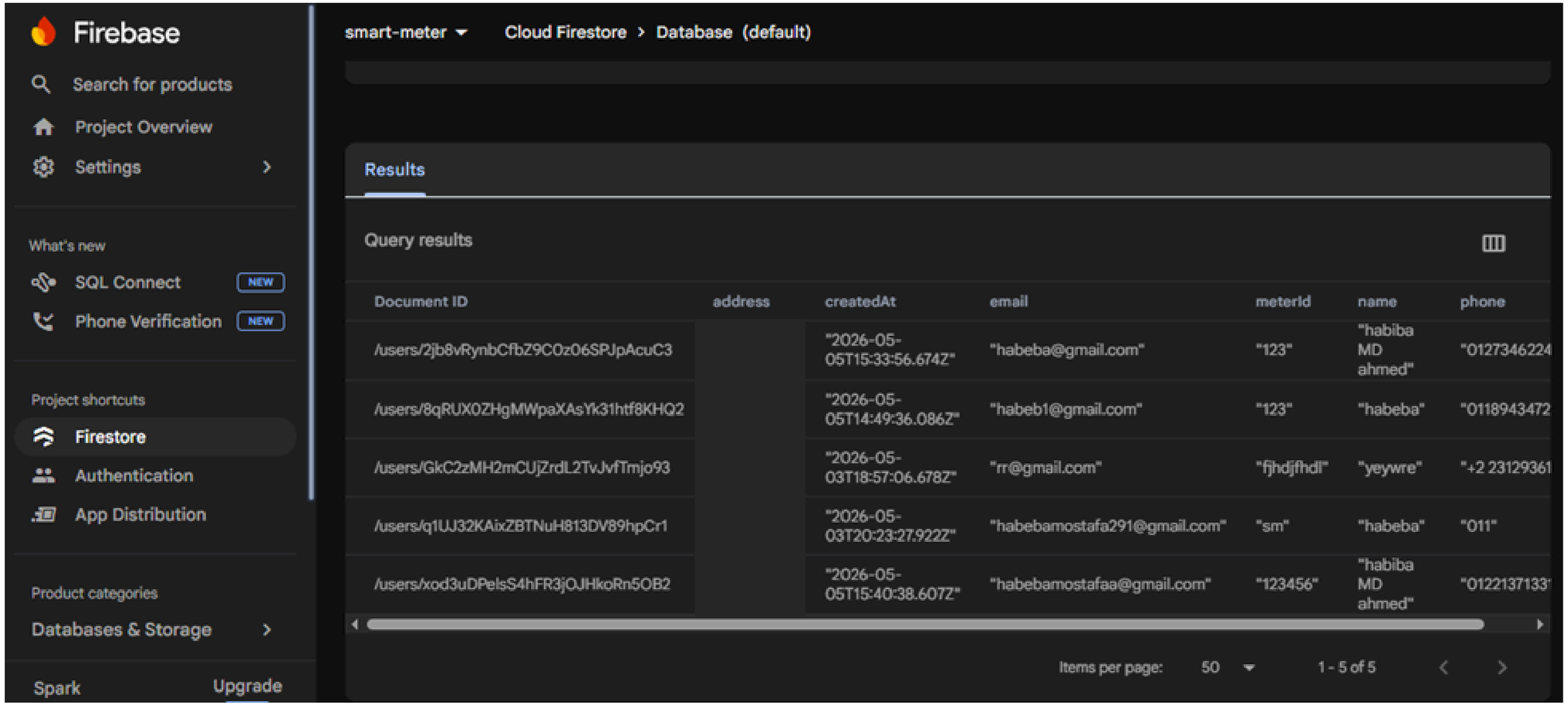
Backend Screens

users



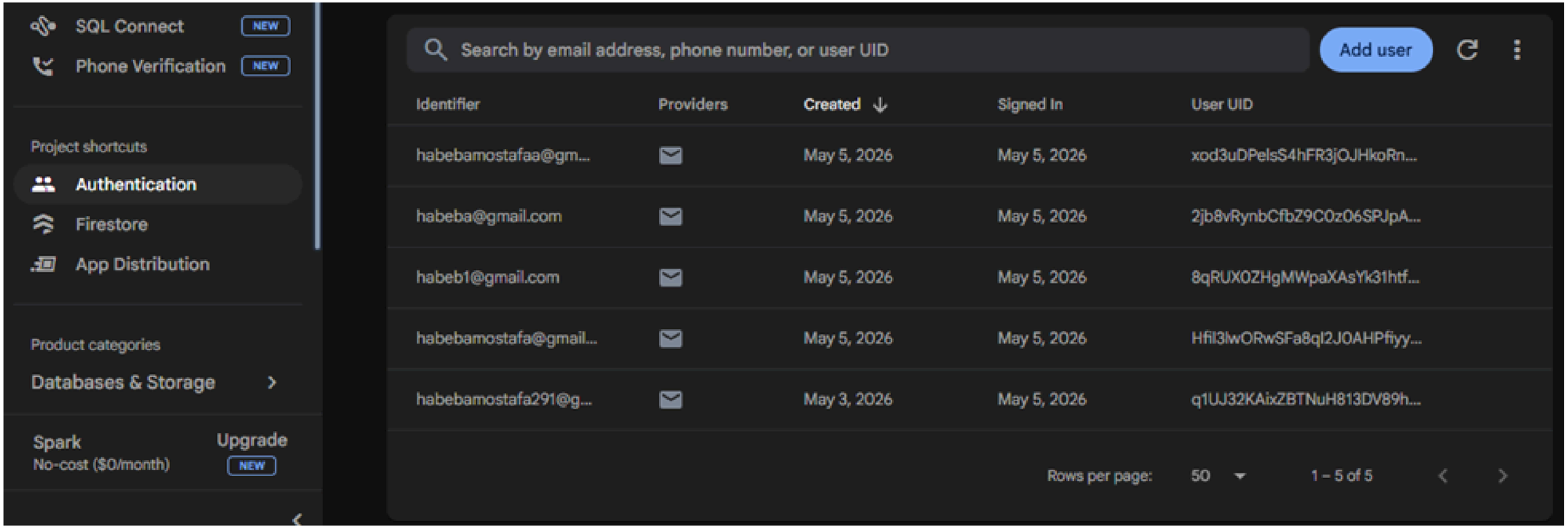
Backend Screens

users



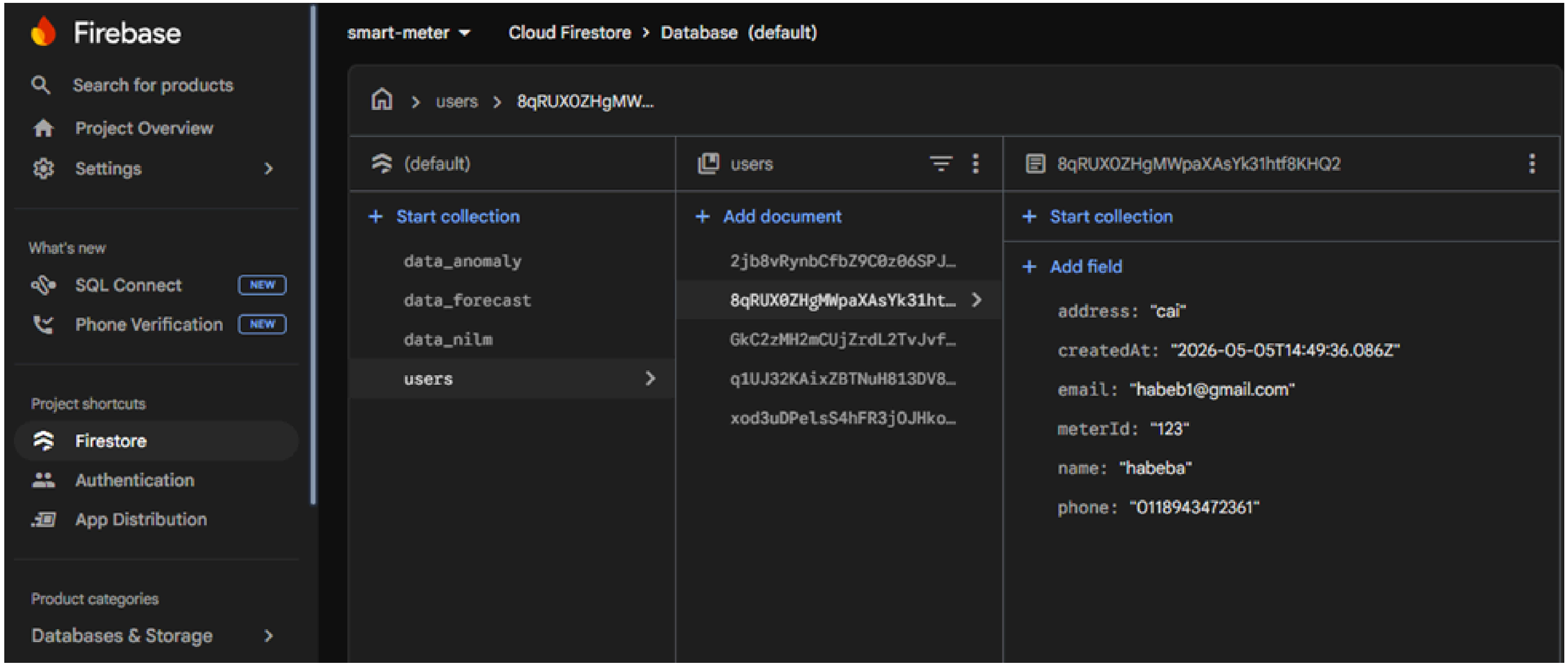
Backend Screens

Authentication

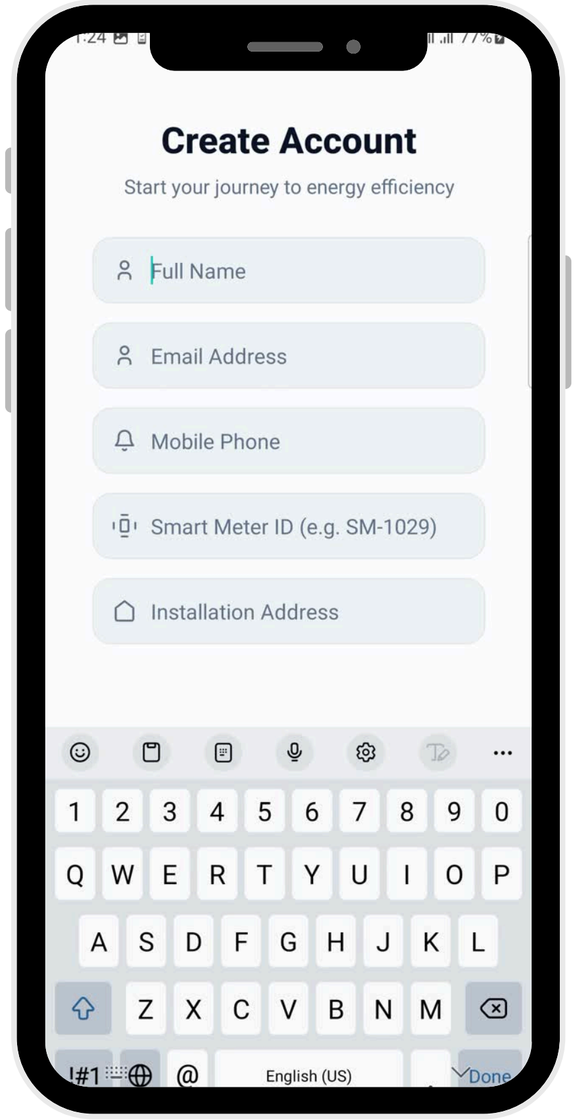


Backend Screens

general overview



RunTime Simulation Demo



Thank You